ROTABLATION-ASSISTED LEFT MAIN PCI

A single centre experience

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BACKGROUND

Calcified left main disease makes PCI hard, yet for some patients PCI is the only option.

Why Indian data matter

- Extensive, multivessel, heavily calcified CAD
- CAD a decade earlier than in the West
- High burden of diabetes and dyslipidemia
- DAPT compliance hit by cost and awareness
- Little rotablation LM-PCI evidence from India

Prospective study design

45

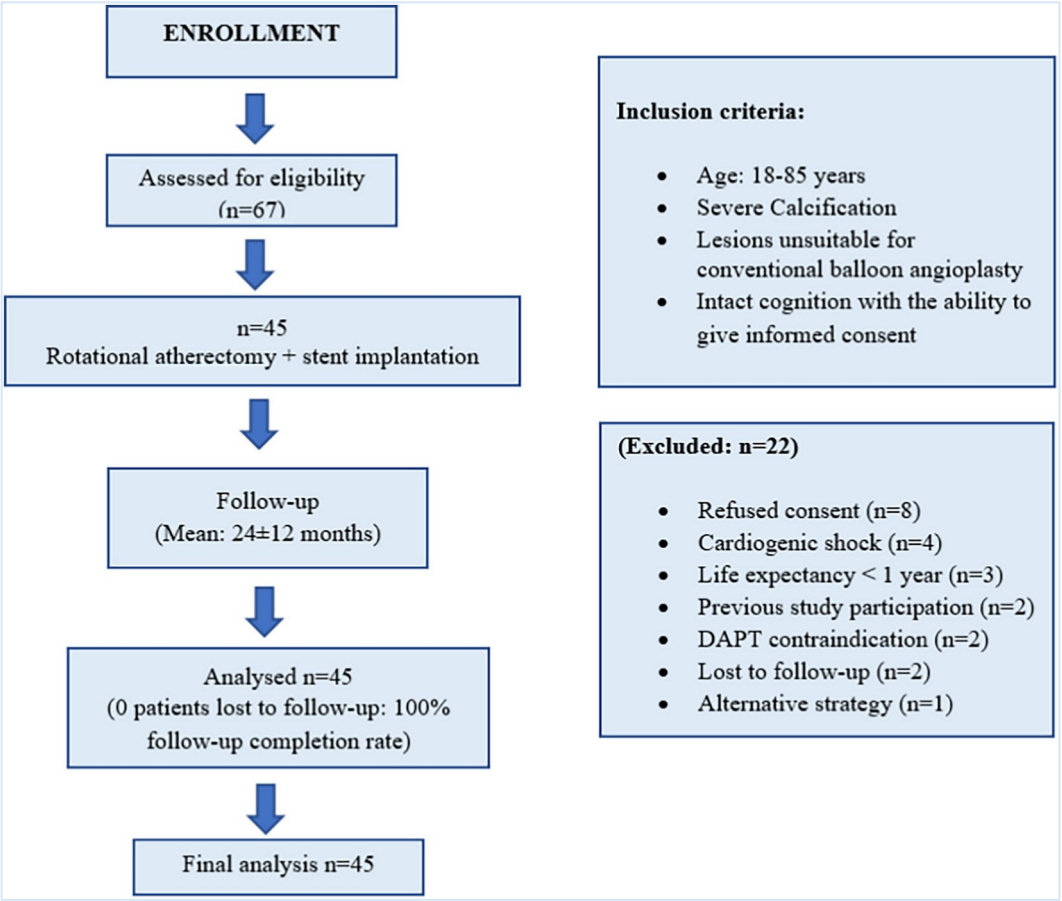
consecutive patients, rota LM-PCI

2020–23

enrolled Jan 2020 – Jan 2023

2 years

follow-up, 100% complete



Patient selection (Fig. 1)

Heart team: SYNTAX >32 advised CABG

PCI by patient choice after counselling

LMCA ≥50% + severe Ca or IVUS Ca score ≥2

Rotablator RA technique

0.7	1.25–2.0	140–180k
target burr : artery ratio	mm burrs; multiple if needed	rpm; burr runs <30 s

Outcome definitions

In-hospital

- Angiographic: <20% residual, TIMI 3
- Procedural: no major complication, death, MI or emergency revasc.

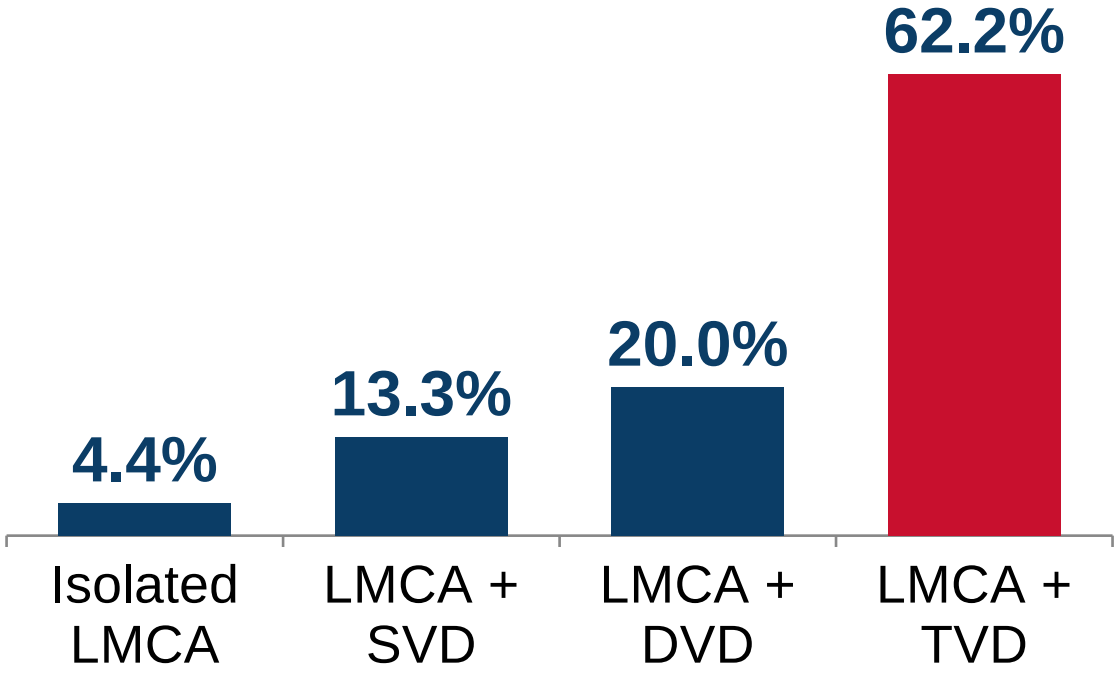
Two-year

- MACE: cardiac death, nonfatal MI, TLR
- Stent thrombosis (ARC), stroke, bleeding (BARC)

Elderly, high-risk cohort

Clinical profile	Value	Risk / history	n (%)
Age (years)	75.2 ± 5.9	Hypertension	38 (84.4%)
Male	31 (68.9%)	Diabetes	25 (55.6%)
LVEF (%)	43.3 ± 8.1	Previous PCI	15 (33.3%)
SYNTAX score	40.1 ± 8.2	Previous CABG	11 (24.4%)
EuroSCORE II	5.4 ± 1.9	NSTEMI	24 (53.3%)

Mostly LM + 3-vessel CAD



86.7%
severe LMCA calcification

62.2%
distal LMCA lesion

Procedural details

Technique	n (%)	Parameter	Value
Femoral access	38 (84.4%)	Max burr, mm	1.50 ± 0.13
7F guide	45 (100%)	Burrs/patient	1.11 ± 0.31
Two-stent	34 (75.6%)	Stent len., mm	22.98 ± 6.54
DK crush	32 (71.1%)	Stent dia., mm	3.49 ± 0.40
IVUS	34 (75.6%)	IABP, n (%)	7 (15.6%)

In-hospital outcomes

100%

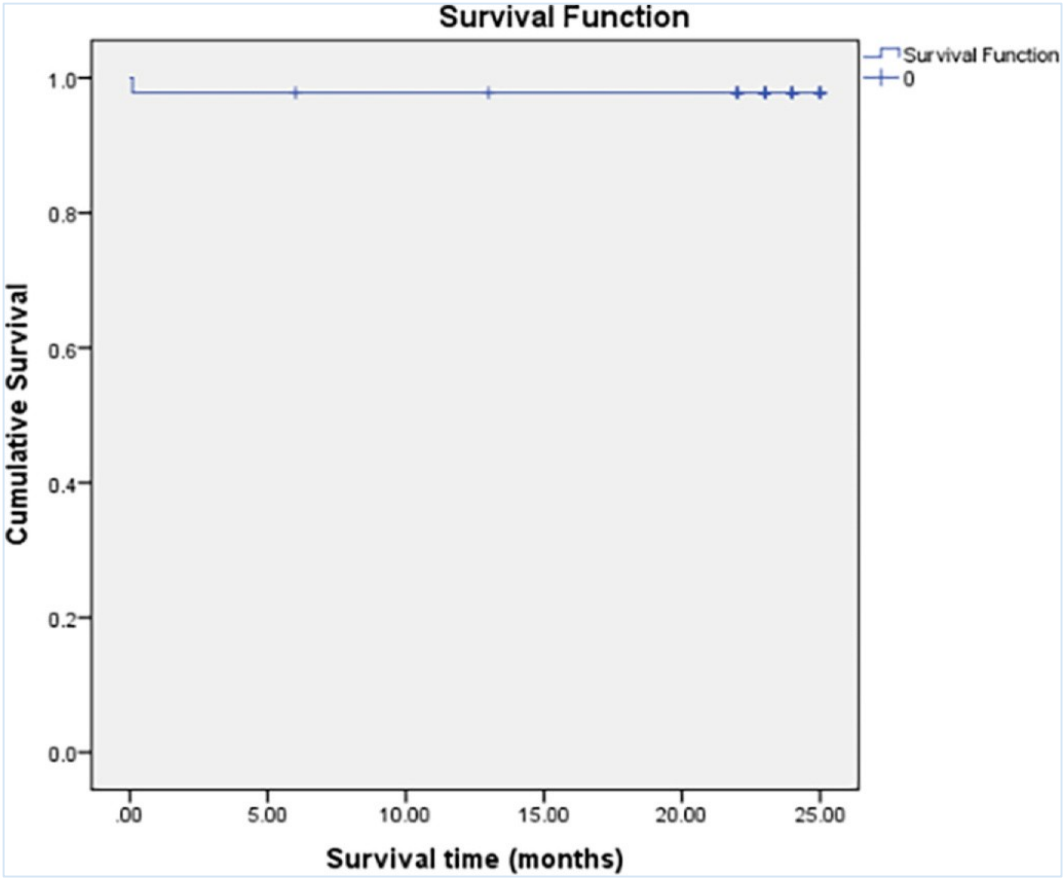
angiographic success (45/45)

97.8%

procedural success (44/45)

2.2%

in-hospital death (1/45)



Two-year outcomes

Overall survival **97.78%**
(Fig. 2, Kaplan-Meier)

MACE-free **93.3%**

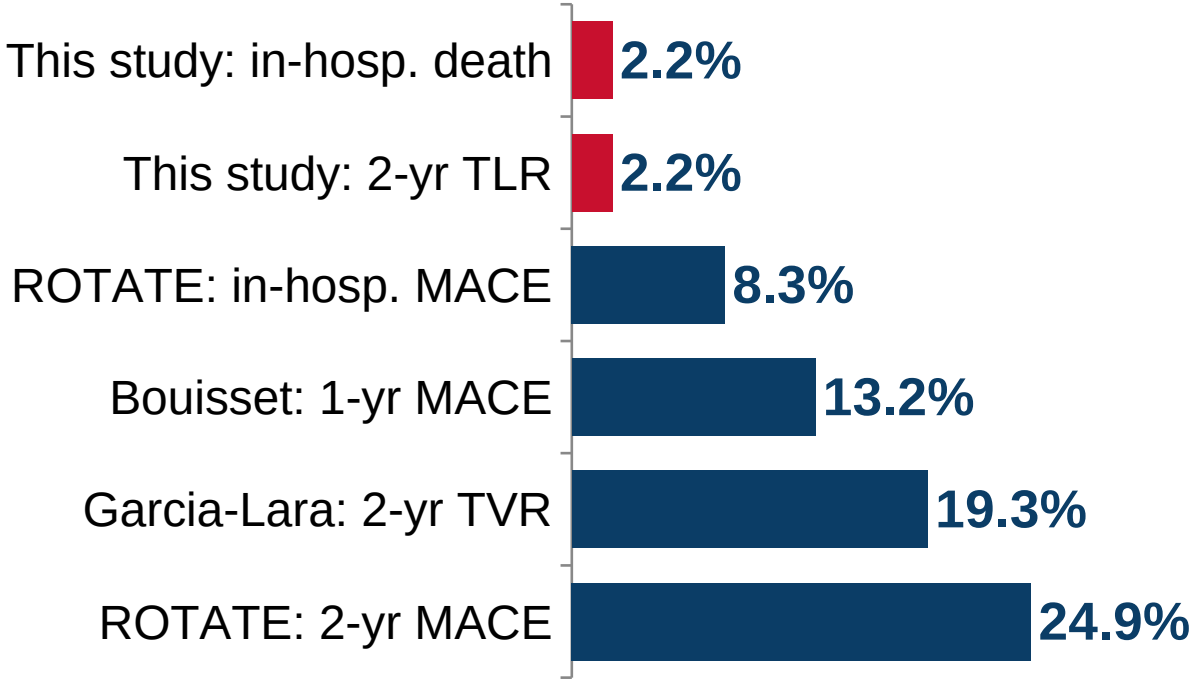
1 death, 1 TLR, 1 TVMI

No bleeding or stroke

Correlation analysis

Variable pair	Pearson r	95% CI	p
Age vs SYNTAX score	0.23	0.02–0.42	0.028
EuroSCORE II vs age	0.52	0.34–0.67	<0.001
LMCA stenosis vs fluoro	0.31	0.11–0.49	0.003
Contrast vs fluoro time	0.38	0.19–0.55	0.001
Burr size vs stent diam.	0.45	0.26–0.61	<0.001
SYNTAX vs stent length	0.28	0.08–0.46	0.006

Endpoints differ by study



Not like-for-like

Garcia-Lara 2-yr cardiac-mortality-free survival

71 ± 7%

Limitations

Design

- Single centre, small sample (n = 45)
- No control group (non-rota PCI or CABG)

Scope

- No systematic angiographic follow-up
- Rotablator RA only: not orbital or IVL

Take-home messages

- Rota-assisted LM-PCI is feasible, safe and effective in severely calcified lesions
- Procedural success 97.8%, with a low rate of periprocedural complications
- Durable two-year outcomes: MACE-free survival 93.3%

THANK YOU

Questions and discussion

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